Formula 1 Race Intelligence: Tableau Project Report

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1. Project Overview

This project is a Tableau analysis of Formula 1 race data: 334 races, 6,464 race records, 19 drivers and 10 teams across 17 seasons (2010 to 2026). It turns the raw data into seven charts, one interactive dashboard and one story that walks through the findings.

The work covers driver performance, team championship points, tyre strategy, race locations, driver reliability (DNFs) and podium results.

2. Why We Made This Project

Race data is large and hard to read as rows in a spreadsheet. Team strategists, broadcasters and regulators each need different answers from it, and a visual dashboard gives those answers quickly.

The project was made to:

- Show how teams and drivers perform over many seasons without reading thousands of rows.
- Help compare a team's points against rivals such as Red Bull Racing and Mercedes.
- Show whether tyre choice, circuit type or failure type changes race outcomes.
- Give a broadcaster ready-made visuals for leaderboards, host countries and podiums.
- Give a regulator evidence on DNF patterns and whether race completion has improved.
- Practise the full Tableau workflow: connecting data, building worksheets, combining them into a dashboard and presenting them as a story.

3. Objectives and User Scenarios

The dashboard serves three users. Each has a different question, and each question is answered by specific worksheets.

Scenario	User	What they need	Worksheets used
1	Mr. Arnold, F1 team performance strategist	Compare his team's points with Red Bull Racing and Mercedes, see how tyre choice affects results, and find the strongest drivers to recruit	Team Trajectory, Tyre Strategy, Driver Leaderboard
2	Ms. Lisa, motorsport broadcast analyst	Quick visuals for leaderboards, host countries and podium counts for on-screen graphics	Driver Leaderboard, Country Participation, Podium Achievements
3	Mr. Gilchrist, FIA technical regulation analyst	Find which failure types cause DNFs and whether race completion has improved across seasons	Driver Reliability, Reliability Trend

4. Dataset

The data came from one Excel workbook, F1_Racing_Project_Dataset.xlsx, with five sheets. Nearly every chart uses the main table, race_results.

Sheet	Rows	Contents
race_results	6,464	One row per driver per race, with 42 fields (circuit, weather, grid and finish position, points, DNF reason, tyre strategy, pit stops)
championship_standings	312	Season totals per driver: points, wins, podiums, DNFs
lap_intelligence	3,814	Lap times, sector times and tyre data
qualifying_results	3,130	Qualifying positions and lap times
circuits	29	Circuit name, country, type and length

The race data covers 17 seasons (2010 to 2026), 334 races, 19 drivers, 10 teams and 21 host countries. Circuits are classed as Permanent, Street or Street/Park.

Fields used most often: Season, Driver, Team, Points, Finish Position, Starting Compound, Circuit Type, Country, Dnf and Dnf Reason.

5. Tools and Techniques Used

Tools

- Tableau Desktop: building worksheets, the dashboard and the story.
- Microsoft Excel: the source data file.
- Mapbox maps (built into Tableau): the country map.

Techniques

- Connected Tableau to the Excel workbook and used the race_results table as the main source.
- Converted Season from a measure to a discrete dimension so it works as a yearly axis.
- Created two calculated fields: Podium Flag and Reliability Rate (formulas in section 6).
- Used filters to remove blank DNF reasons (finishers) and to limit Team Trajectory to selected teams.
- Used marks and encodings: color for teams and failure types, size and color for race counts, and labels for exact values.
- Applied a dark map style and a dark dashboard theme with consistent colors, borders and padding.
- Added dashboard interactivity so that selecting a chart filters the others.
- Built a story with a caption on each point.

6. Project Summary in Numbers

Item	Count
Worksheets (charts)	7
Dashboards	1
Stories	1 (6 story points)

Item	Count
Calculated fields	2
Data source	1 Excel workbook (5 sheets, mainly race_results)
Race records analysed	6,464
Seasons covered	17 (2010 to 2026)

The two calculated fields:

- **Podium Flag** marks a podium finish: `IF [Finish Position] <= 3 THEN 1 ELSE 0 END`
- **Reliability Rate** is the share of race entries that finished: `1 - SUM([Dnf]) / COUNT([Race Id])`

7. Worksheets

Seven worksheets were built, each answering one question.

#	Worksheet	Chart type	Fields used	Question answered
1	Driver Leaderboard	Sorted horizontal bar	Driver, Points	Which drivers scored the most points overall?
2	Team Trajectory	Line chart	Season, Points, Team (filtered to 3 teams)	How do teams' points change season by season?
3	Tyre Strategy	Bars by circuit type	Starting Compound, Circuit Type, Average Finish Position	Does the starting tyre change finishing position on different circuits?
4	Country Participation	Symbol map	Country, Count Distinct of Race Id	Which countries host the most races?
5	Driver Reliability	Stacked bar	Driver, Dnf, Dnf Reason	Which drivers retire most, and why?

#	Worksheet	Chart type	Fields used	Question answered
6	Reliability Trend	Line chart	Season, Reliability Rate	Has race completion improved over the seasons?
7	Podium Achievements	Bar chart	Driver, Podium Flag, Team	Who has the most podium finishes?

8. Dashboard and Story

Dashboard. One dashboard, Formula 1 Race Intelligence Dashboard, combines all seven worksheets on one screen. Driver Leaderboard and Tyre Strategy sit at the top left, Country Participation and Team Trajectory beside them, and Driver Reliability, Reliability Trend and Podium Achievements along the bottom. Charts act as filters: selecting a driver in the leaderboard updates the other charts, and selecting it again clears the filter. A shared legend shows team and failure-type colors.

Story. One story, F1 Race Intelligence, presents the findings in six points, each with a caption:

1. Team Trajectory: how Mercedes and Red Bull compare over time.
2. Tyre Strategy: the starting tyre makes only a small difference to finishing position.
3. Driver Leaderboard: the top drivers by total points.
4. Podium Achievements: podium counts follow the points leaderboard.
5. Driver Reliability: DNFs are spread across all failure types.
6. Reliability Trend: race completion rate by season.

9. Key Insights

Max Verstappen is the strongest driver in this dataset, and race reliability and tyre choice change little between seasons or compounds.

- **Top drivers.** Max Verstappen leads total points and has the most podiums (187). Lewis Hamilton is second on both. Both are the benchmark for a team recruiting talent.
- **Team trajectory.** Mercedes peaked at about 810 points around 2019. Red Bull Racing's points swing more from season to season, with peaks near 2013 and 2023. A mid-tier team must close a gap of several hundred points to match either.

- **Tyre strategy.** Average finish position stays between about 9.9 and 11.0 for every starting compound on every circuit type. Starting tyre alone is not a strong predictor of results here.
- **DNFs.** 396 of 6,464 race entries (6.1%) ended in a DNF. The six causes are close in size: Engine (73), Hydraulics (70), Mechanical (67), Brake failure (64), Gearbox (62) and Collision (60). No single failure type dominates.
- **Reliability trend.** The completion rate stays between about 92% and 96% in every season, so the data shows no clear improvement over time.
- **Race locations.** Races were held in 21 countries across the 17 seasons.

10. Design Choices, Limitations and Conclusion

Design choices

- A dark theme with thin panel borders and one red accent colour, so the dashboard looks like a racing control screen.
- Insight-led chart titles and short story captions, so each chart states its point.
- Colour used for meaning only: teams, failure types and tyre compounds.

Limitations

- The dataset has no driver nationality field, so the nationality talent-pool analysis in the original brief was not built. Race host country is shown instead.
- The data looks simulated rather than taken from real F1 history (for example, Max Verstappen appears with Red Bull Racing in 2010). The findings describe this dataset only.
- Carlos Sainz has 668 records, twice the 334 of most drivers, so his totals are higher than a like-for-like comparison would give.
- The lap_intelligence, qualifying_results and circuits sheets were not used in the charts.

Conclusion

The project answers the questions of all three user scenarios in one dashboard and one story: who performs best, how teams compare, whether tyre choice matters, and how reliable the cars are. Further work could add driver nationality data, use the lap and qualifying tables for pace analysis, and test reliability against regulation changes by year.